

What to Fix and What to Estimate in Bayesian Structural Models

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Structural models often calibrate some parameters and estimate others, yet this partition is typically chosen informally. We propose a partition-selection criterion that selects, among admissible partitions, the one minimizing the worst-case local bias of the posterior mean induced by calibration errors. We derive an exact finite-sample sensitivity formula, avoiding Bernstein–von Mises approximations. The Bayesian rule is asymptotically equivalent to its minimum-distance counterpart, but Monte Carlo and Smets–Wouters evidence shows substantial finite-sample differences. The method is straightforward to implement and available for MATLAB/Dynare.

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JEL CLASSIFICATION. C10, C12, C16.

1. INTRODUCTION

Over the past two decades, Bayesian estimation of structural models, particularly dynamic stochastic general equilibrium (DSGE) models, has become a standard approach in macroeconomics. A common practice is to calibrate a subset of structural parameters—that is, to hold them fixed at prespecified values—and

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estimate the remaining parameters.¹ Calibration may be motivated by identification problems, computational constraints, or credible external information about particular parameters. Yet the calibration–estimation partition is often inherited from earlier work or chosen informally.

This paper treats this choice as a partition-selection problem. For each admissible partition, we measure the local sensitivity of the posterior mean, of an object of interest of the paper, to perturbations in the calibrated parameters. The method selects the admissible partition with the minimum sensitivity. By minimizing the sensitivity we actually minimize the worst-case bias due to local miscalibration of the fixed parameters.

As shown in the numerical example and empirical application below, and in the applications of [Alegre \(2026\)](#), the choice of which parameters to calibrate and which to estimate can have substantial implications for the credibility of the resulting conclusions. To illustrate, consider the New Keynesian model of [An and Schorfheide \(2007\)](#). In our Monte Carlo exercise, under a (5%) miscalibration, some partitions produce changes in the posterior mean of the object of interest of more than (18%), whereas the partition selected by our methodology reduces this effect to approximately to (2.5%). Thus, substantial gains in robustness can be achieved simply by choosing the calibration–estimation partition more carefully. The methodology is also straightforward to implement. We provide a MATLAB/Dynare package for DSGE applications; see [Section 4](#).

A closely related paper is [Alegre \(2026\)](#), which develops an analogous partition-selection method in a minimum-distance framework. [Alegre \(2026\)](#) selects the partition that minimizes the worst-case local bias of the object of interest, whereas in the present paper we minimize the worst-case local bias of its posterior mean. In [Section 4](#), we show that the two partition-selection procedures are

¹Throughout the paper, a parameter is *calibrated* if it is held fixed during estimation. This usage is narrower than in some applied and econometric work, where calibration means choosing parameters to match empirical moments, i.e., minimum-distance estimation; see [Cocci and Plagborg-Møller \(2021\)](#). We refer to the latter as estimation.

asymptotically equivalent under regularity conditions. In finite samples, however, the Monte Carlo simulation and empirical application show that the two methods can select markedly different partitions.

A second contribution is to provide a closed-form expression for the local sensitivity of the posterior mean of an estimated parameter, or more generally of an object of interest, to miscalibration of the fixed parameters. This contribution is closely related to [Iskrev \(2019\)](#), who develops a sensitivity measure for posterior means with respect to calibrated parameters based on a Bernstein–von Mises asymptotic approximation. In contrast, we derive the exact finite-sample sensitivity of the posterior mean, without relying on this asymptotic approximation.

To implement the methodology, we require a baseline parameter vector, typically obtained from a first-stage estimation of the model; see [Section 4](#). The robust partition minimizes the worst-case bias induced by miscalibrations around this baseline, so the procedure is inherently local.

A second key input is the set of normalization ranges for the calibrated parameters. These ranges place calibration errors in comparable units and incorporate external information about what constitutes a plausible miscalibration. This differs from [Iskrev \(2019\)](#), whose normalization uses standard deviations from the unrestricted asymptotic posterior and therefore requires the full structural parameter vector to be locally identified. Since our procedure is designed to accommodate settings in which full identification may fail, we instead construct the normalization from external information. This is particularly natural in macroeconomic applications, where informative priors are often available for structural parameters. Different normalization ranges may therefore lead to different partition selections, reflecting different views about plausible calibration errors.

[Section 2](#) defines the framework and robust partition. [Section 3](#) derives the sensitivity criterion. [Section 4](#) discusses implementation and the reference point. [Sections 6](#) and [7](#) present the Monte Carlo simulation and the Smets–Wouters application. [Section 8](#) concludes.

2. BAYESIAN FRAMEWORK AND ROBUST PARTITION

This section defines the worst-case bias criterion used to compare alternative calibration–estimation partitions. A partition determines which parameters are estimated and which are fixed. For each admissible partition, we compute the posterior distribution at a common reference point and evaluate its robustness to local miscalibration of the fixed parameters.

We observe $Y_{1:T} = (y_1, \dots, y_T)$, generated by a strictly stationary and ergodic stochastic process. The structural model is indexed by $\eta \in \mathcal{N} \subset \mathbb{R}^{d_\eta}$, and the likelihood is $f(Y_{1:T} \mid \eta)$.

A partition is an index set $S \subseteq \{1, \dots, d_\eta\}$. The parameters indexed by S are estimated, while the parameters in S^c are fixed. We write

$$\alpha := (\eta_j)_{j \in S}, \quad \beta := (\eta_j)_{j \in S^c}, \quad \eta = (\alpha', \beta')'.$$

where α denotes the estimated parameters and β denotes the fixed parameters. All objects below are partition-specific. To keep notation light, we suppress the dependence on S whenever this causes no ambiguity.

Given a prior $\pi(\alpha)$ and a fixed value β , the posterior distribution of the estimated parameters is

$$\pi_T(\alpha \mid \beta, Y_{1:T}) \propto f(Y_{1:T} \mid \alpha, \beta; S) \pi(\alpha).$$

The object of interest is

$$\gamma = \Gamma(\alpha, \beta; S) \in \mathbb{R}^{d_\gamma}.$$

Examples include a structural parameter, a counterfactual impulse response, a welfare measure, or an average treatment effect. The posterior mean of the object of interest, when the fixed block is set equal to β , is

$$\gamma_T(\beta; S) := \int \Gamma(\alpha, \beta; S) \pi_T(\alpha \mid \beta, Y_{1:T}) d\alpha. \tag{1}$$

2.1 Posterior Mean Worst-Case Bias

The procedure is conditional on a reference point $\eta_T^* \in \mathcal{N}$. For a partition S , write

$$\alpha_T^* := (\eta_{T,j}^*)_{j \in S}, \quad \beta_T^* := (\eta_{T,j}^*)_{j \in S^c}.$$

The reference point is the baseline value around which local calibration errors are evaluated. The robust partition defined below is conditional on η_T^* . Section 4.1 discusses how to obtain this reference point and states the regularity conditions it must satisfy.

Calibration errors must be measured in comparable units. For each partition, define

$$\Sigma_S := \text{diag}((\Delta_j)_{j \in S^c}), \quad \Delta_j := \eta_{j,\max} - \eta_{j,\min},$$

where $(\eta_{j,\min}, \eta_{j,\max})$ is a plausible range for parameter j . These ranges may come from parameter-space restrictions, prior quantiles, or external estimates.

For a scalar $\epsilon > 0$, define the local miscalibration neighborhood

$$\mathcal{B}_\epsilon(\beta_T^*; S) := \left\{ \beta_0 \in \mathcal{B}_S : \|\Sigma_S^{-1}(\beta_0 - \beta_T^*)\|_2 \leq \epsilon \sqrt{|S^c|} \right\}. \quad (2)$$

The factor $\sqrt{|S^c|}$ takes into account the number of calibrated parameters is used in the partition. The robust posterior worst-case bias of partition S is defined by

$$\mathcal{R}_T(S; \eta_T^*, \Sigma_S, \epsilon) := \sup_{\beta_0 \in \mathcal{B}_\epsilon(\beta_T^*; S)} \|\gamma_T(\beta_T^*; S) - \gamma_T(\beta_0; S)\|_2. \quad (3)$$

Expression (3) is the worst-case squared bias generated by local miscalibration of the fixed block. When $S^c = \emptyset$, the bias term is zero by convention.

In the next subsection, we use this to select the minimum worst-case bias partition from a particular class of admissible partitions, $\mathcal{S}_{AD}$.

2.2 Admissible Set and Robust Partition

The worst-case bias is minimized only over partitions for which some regularity conditions are satisfied. Let $\mathcal{H}$ denote any hard restrictions imposed by the researcher, such as parameters that must always be estimated or always be fixed. If there are no such restrictions, take $\mathcal{H} = 2^{\{1, \dots, d_\eta\}}$.

ASSUMPTION 1 (Admissibility). A partition $S \in \mathcal{H}$ is admissible at the reference point η_T^* if:

1. *Non-triviality*. The object of interest is not locally determined only by the fixed block. In particular, $\Gamma(\alpha, \beta; S)$ is not locally constant in α at (α_T^*, β_T^*) .
2. *Conditional local identification*. With β_T^* fixed, the estimated block α is locally identified.

The admissible set is

$$\mathcal{S}_{\text{AD}}(\eta_T^*) := \{S \in \mathcal{H} : S \text{ satisfies Assumption 1}\}.$$

In applications, conditional local identification can be checked using standard rank or Fisher-information diagnostics.

DEFINITION 1 (Robust partition). Conditional on the reference point η_T^* , the normalization matrices $\{\Sigma_S\}_{S \in \mathcal{S}_{\text{AD}}}$, and the local radius ϵ , a robust partition is any solution to

$$S_T^*(\eta_T^*, \{\Sigma_S\}, \epsilon) \in \arg \min_{S \in \mathcal{S}_{\text{AD}}(\eta_T^*)} \mathcal{R}_T(S; \eta_T^*, \Sigma_S, \epsilon). \quad (4)$$

3. SENSITIVITY STATISTIC TO APPROXIMATE THE WORST-CASE BIAS

This section shows how the worst-case bias in (3) can be approximated by a sensitivity statistic. In particular, for small ϵ ,

$$\mathcal{R}_T(S; \eta_T^*, \Sigma_S, \epsilon) = \epsilon K_{T,S} + o(\epsilon),$$

where $K_{T,S}$ measures the local sensitivity of $\gamma_T(\beta_T^*; S)$ to the calibrated parameters β_T^* . This allows us to avoid solving the maximization problem in the worst-case bias definition.

We first develop closed-form solutions for $\partial\alpha/\partial\beta$ and $\partial\gamma/\partial\beta$. In next subsection 3.1, we construct $K_{T,S}$ and relate it to the worst-case bias. The log-likelihood function and the score functions are defined as follows.

$$\ell_T(\alpha, \beta; S) = \log f(Y_{1:T} \mid \alpha, \beta; S), \quad \nabla_{\eta} \ell_T(\eta; S) := \frac{\partial \ell_T(\eta; S)}{\partial \eta}. \quad (5)$$

Denote by $\nabla_\alpha \ell_T(\alpha, \beta; S)$ and $\nabla_\beta \ell_T(\alpha, \beta; S)$ the corresponding columns of $\nabla_\eta \ell_T(\eta; S)$.

The mean of the posterior distribution is:

$$\alpha_T(\beta) := \int \alpha \pi_T(\alpha | Y_{1:T}, \beta) d\alpha. \quad (6)$$

Suppose that the conditions to differentiate (6) under the integral sign are satisfied, the derivative of $\alpha_T(\beta; S)$ with respect to β is:

$$\begin{aligned} \frac{\partial \alpha_T(\beta; S)}{\partial \beta} &= \int \alpha \frac{\partial}{\partial \beta} \pi_T(\alpha | Y_{1:T}, \beta) d\alpha \\ &= \int \alpha (\nabla_\beta \ell_T - E_{\pi_T}[\nabla_\beta \ell_T]) \pi_T(\alpha | Y_{1:T}, \beta) d\alpha \\ &= \text{Cov}_{\pi_T}(\alpha, \nabla_\beta \ell_T). \end{aligned} \quad (7)$$

To follow all intermediate steps, check Appendix B. With the posterior distribution, $\pi_T(\alpha | Y_{1:T}, \beta)$, we can easily obtain $\text{Cov}_{\pi_T}(\alpha, \nabla_\beta \ell_T)$ without re-estimating the model. Doing the same exercise for the object of interest:

$$\begin{aligned} \frac{\partial \gamma_T(\beta; S)}{\partial \beta} &= \int \frac{\partial \Gamma(\alpha, \beta)}{\partial \beta} \pi_T(\alpha | Y_{1:T}, \beta) d\alpha + \int \gamma \frac{\partial \pi_T(\alpha | Y_{1:T}, \beta)}{\partial \beta} d\alpha \\ &= E_{\pi_T}\left(\frac{\partial \Gamma(\alpha, \beta)}{\partial \beta}\right) + \text{Cov}_{\pi_T}(\gamma, \nabla_\beta \ell_T). \end{aligned} \quad (8)$$

The closest sensitivity statistic is Iskrev (2019), who studies the sensitivity of posterior means of estimated parameters to perturbations in calibrated parameters; see Section 5.1. Unlike Iskrev's asymptotic statistic, equations (7)–(8) provide the closed form solution of the finite-sample sensitivity. For transparency, they can also be range-normalized to report the percentage response of an estimated parameter, or of the object of interest, to a perturbation equal to one percent of a calibrated parameter's plausible range; see Appendix A.1.

3.1 Sensitivity Statistic for Partitions

Note that $\partial \gamma_T(\beta; S)/\partial \beta$ depends on the specific units of β . Consequently, it cannot be used to compare the sensitivity across different partitions. To eliminate this unit dependence, we normalize the derivative using the range normalization matrix, Σ_S , defined above.

DEFINITION 2. The *sensitivity statistic for the posterior mean* is defined by

$$K_{T,S} := \sqrt{|S^c|} \left\| \frac{\partial \gamma_T(\beta_T^*; S)}{\partial \beta} \Sigma_S \right\|_2. \quad (9)$$

An additional advantage of using the normalization matrix Σ_S is that it allows us to incorporate information into the sensitivity analysis. The following example illustrates this.

EXAMPLE. Consider the simple case where the object of interest and the fixed parameters are scalars. Suppose $S^c = \{j\}$. The sensitivity statistic is:

$$K_{T,S} = \underbrace{\left| \frac{\partial \gamma_T(\beta_T^*; S)}{\partial \beta} \right|}_{\text{Model's implied sensitivity}} \underbrace{(\eta_{j,\max} - \eta_{j,\min})}_{\text{Additional Information}}. \quad (10)$$

Thus, $K_{T,S}$ can be view as a combination of how much we know about the fixed parameter, and how sensitive is the object of interest to miscalibrations.

LEMMA 1. *The statistic $K_{T,S}$ determines the first-order worst-case local bias. In particular,*

$$R_T(S; \eta_T^*, \Sigma_S, \epsilon) = \epsilon K_{T,S} + o(\epsilon). \quad (11)$$

PROOF. See Appendix B. □

ASSUMPTION 2. The problem $\min_{S \in S_{AD}(\eta_T^*)} K_{T,S}$ has a unique solution S_T^* .

LEMMA 2. *Suppose Assumption 2 holds and the expansion (11) is valid uniformly over $S \in S_{AD}(\eta_T^*)$. Then there exists $0 < E \leq \bar{\epsilon}$ such that, for every $0 < \epsilon \leq E$,*

$$\arg \min_{S \in S_{AD}(\eta_T^*)} R_T(S; \eta_T^*, \Sigma_S, \epsilon) = \arg \min_{S \in S_{AD}(\eta_T^*)} K_{T,S} := S_T^*.$$

PROOF. See Appendix B. □

Lemma 2 shows that, for sufficiently small local miscalibrations, the robust partition can be obtained by minimizing the sensitivity statistic $K_{T,S}$

4. IMPLEMENTATION: ALGORITHM AND REFERENCE POINT

This section describes how to implement the partition selection criterion developed above.

Algorithm for Selecting a Locally Least-Sensitive Partition

Step 1. Take as inputs a reference parameter vector η_T^* , miscalibration size ϵ , a set of hard restrictions on admissible partitions $\mathcal{H}$, a family of normalization matrix Σ_S for each partition in $\mathcal{H}$, a prior distribution for each estimated parameter estimated at least in some partition in $\mathcal{H}$, and a threshold for rank estimation τ_t .

Step 2. Compute the set of admissible partitions, $\mathcal{S}_{AD}(\eta_T^*)$:

Step 2.1. Start with the set $\mathcal{H}$.

Step 2.2. From the set $\mathcal{H}$, rule out any S for which the object of interest is entirely explained by the calibrated parameters.

Step 2.3. For every S in the remaining set, compute the estimated Fisher information matrix:

$$\hat{\mathcal{I}}_{\alpha\alpha,S}(\eta_T^*) := -\frac{1}{T} \frac{\partial^2 \ell_T(\alpha, \beta; S)}{\partial \alpha \partial \alpha'} \Big|_{(\alpha, \beta) = \eta_T^*}.$$

Step 2.4. Compute the minimum singular value of the estimated information matrix,

$$\sigma_{\min}(\hat{\mathcal{I}}_{\alpha\alpha,S}(\eta_T^*)),$$

and keep S in $\hat{\mathcal{S}}_{AD}(\eta_T^*)$ if $\sigma_{\min}(\hat{\mathcal{I}}_{\alpha\alpha,S}(\eta_T^*)) > \tau_t$.

Step 3. Compute the approximate risk for every $S \in \hat{\mathcal{S}}_{AD}(\eta_T^*)$:

Step 3.1. Fix $\beta_T^* = (\eta_{T,j}^*)_{j \in S^c}$.

Step 3.2. Compute the posterior distribution $\pi_T(\alpha \mid \beta_T^*, Y_{1:T}; S)$.

Step 3.3. Compute the derivative

$$\frac{\partial \gamma_T(\beta_T^*; S)}{\partial \beta} = \mathbb{E}_{\pi_T} \left[\frac{\partial \Gamma(\alpha, \beta_T^*)}{\partial \beta} \right] + \text{Cov}_{\pi_T}(\gamma, \nabla_{\beta} \ell_T).$$

Step 3.4. With the normalization matrix Σ_S , compute

$$K_{T,S} := \sqrt{|S^c|} \left\| \frac{\partial \gamma_T(\beta_T^*; S)}{\partial \beta} \Sigma_S \right\|_2.$$

Step 4. Solve $\min_{S \in \widehat{\mathcal{S}}_{\text{AD}}(\eta_T^*)} K_{T,S}^2$.

The implemented method is conditional on several choices: the reference point η_T^* , the normalization matrices $\{\Sigma_S\}$, the tuning parameter τ_t used for rank estimation, and the prior distributions for any parameters estimated under at least one partition. Among these inputs, the reference point plays a central role. It provides the common point around which local calibration errors are evaluated, and therefore makes the calibration risk comparable across different partitions. The next subsection discusses this choice in detail. It states the conditions under which such comparability is obtained and proposes a practical way to construct a reference point satisfying those conditions.

To facilitate application of the procedure, this paper is accompanied by a MATLAB implementation designed to work with Dynare. For a user-specified set of admissible parameter partitions and a scalar or vector object of interest, the software computes the finite-sample Bayesian and asymptotic-MLE sensitivity criteria and reports the least-sensitive partition within that set. The repository contains the source code, worked examples, numerical-validation scripts, and usage instructions.²

4.1 Reference Point Implementation

To compare different partitions, the posterior distributions must be evaluated around a common population object. The posterior distribution of the estimated block should concentrate around the corresponding S -coordinates of the reference point. Under the usual Bernstein–von Mises conditions, this comparability is obtained if the reference point satisfies the following condition.

ASSUMPTION 3 (Reference Point).

$$\eta_T^* \rightarrow_p \eta^* \in \mathcal{N}_0, \quad \mathcal{N}_0 := \{\eta \in \mathcal{N} : P_0 = P_\eta\}, \quad (12)$$

²The research code is available at <https://joanalegre.github.io/index.html>. The version accompanying this paper is release version 0.1.0, tested with MATLAB R2025b and Dynare 7.1. The interface is under active development.

where P_0 denote the true law of the stochastic process $\{Y_t\}_{t \geq 1}$, and P_η denote the law implied by the full model at η .

The estimated reference point η_T^* can be estimated by either MLE or bayesian estimation, using a first-stage calibration (S_1, β_1) . In the case of MLE first-stage estimation:

$$\eta_T^* := (\hat{\alpha}, \beta_1), \quad \text{with} \quad \hat{\alpha} = \arg \max_{\alpha \in \mathcal{A}} \left\{ \frac{1}{T} \ell_T(\alpha, \beta_1) \right\}. \quad (13)$$

In the case of Bayesian first-stage estimation, $\hat{\alpha}$ may instead be taken as the posterior mean of $\pi_T(\alpha \mid Y_{1:T}, \beta_1; S_1)$, or as the posterior mode. Under the usual Bernstein–von Mises conditions, the posterior concentrates around the maximum likelihood estimator, so either choice yields the same limiting reference point.

A primitive condition for Assumption 3 is correct specification of the restricted model, i.e, for some $\alpha \in \mathcal{A}$

$$P_0 = P_{\alpha, \beta_1}. \quad (14)$$

When correct specification of the restricted model is too strong, Appendix A.2 provides an alternative construction requiring only correct specification of the full model, $\mathcal{N}_0 \neq \emptyset$.

5. RELATION TO THE LITERATURE ON SENSITIVITY AND PARTITION SELECTION

The paper’s main contribution is a procedure for selecting which structural parameters to estimate and which to fix by minimizing calibration risk over admissible partitions. A key building block of the procedure is a finite-sample formula for the sensitivity of posterior means to perturbations in the fixed parameters. This section first relates the sensitivity formula to existing measures and then compares the proposed partition-selection procedure with related methods.

5.1 Discussion of Sensitivity

The closest papers to this one in terms of $\partial \alpha_T(\beta) / \partial \beta$ are [Iskrev \(2019\)](#) and [Jørgensen \(2023\)](#). We first discuss the relationship with [Jørgensen \(2023\)](#).

Jørgensen (2023) develops a sensitivity statistic for estimated parameters with respect to fixed parameters in a Generalized Method of Moments environment. In the MLE case, let $\hat{\alpha}(\beta_T^*)$ denote the estimator obtained when β_T^* is calibrated. The sensitivity of the estimated parameters to the calibrated parameters is

$$\left. \frac{\partial \hat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} := \hat{I}_{\alpha, \alpha}^{-1} \hat{I}_{\alpha, \beta},$$

where

$$\hat{I}_{\alpha, \alpha} := -\frac{1}{T} \frac{\partial^2 \ell_T(\alpha, \beta)}{\partial \alpha \partial \alpha'} \Big|_{(\alpha, \beta) = (\hat{\alpha}(\beta_T^*), \beta_T^*)}, \quad \hat{I}_{\alpha, \beta} := \frac{1}{T} \frac{\partial^2 \ell_T(\alpha, \beta)}{\partial \alpha \partial \beta'} \Big|_{(\alpha, \beta) = (\hat{\alpha}(\beta_T^*), \beta_T^*)}.$$

Theorem 3 in Appendix A.3 develops some regularity conditions for

$$\left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} - \left. \frac{\partial \hat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} = o_p(1),$$

In the DSGE literature, Iskrev (2010) also studies the sensitivity of the posterior mean of estimated parameters with respect to calibrated parameters. That paper uses a Bernstein–von Mises approximation to the unrestricted posterior distribution of the full parameter vector η and consequently approximates $\partial \alpha_T(\beta) / \partial \beta$ by the corresponding MLE sensitivity. Thus, the unnormalized derivative in Iskrev (2019) coincides algebraically with the MLE sensitivity in Jørgensen (2023).

There are two main differences with our approach. First, equation (7) gives the exact finite-sample derivative of the posterior mean, whereas Iskrev’s derivative is based on the Bernstein–von Mises approximation. Second, Iskrev normalizes perturbations in the calibrated parameters by their standard deviations under the unrestricted asymptotic posterior. This provides a data-driven normalization, whereas Σ_S is constructed from external information.

Because Iskrev’s normalization relies on the covariance matrix of the unrestricted asymptotic posterior, it requires the full parameter vector η to be locally identified. Our procedure instead requires only conditional local identification of the estimated block under each candidate partition. This distinction is important because lack of full identification is itself a central reason for fixing structural parameters.

Sections 6 and 7 show that the Bernstein–von Mises approximation can be inaccurate in small samples and can therefore lead to a different choice of calibration–estimation partition.

For the sensitivity statistic $K_{T,S}$, the closest reference is Alegre (2026), which works in a Minimum Distance framework. In the MLE case, its statistic is

$$\hat{K}_S := \sqrt{|S^c|} \|\hat{D}_{\gamma,\beta} \Sigma_S\|_2, \quad (15)$$

where

$$\hat{D}_{\gamma,\beta} := \frac{\partial \Gamma(\alpha_T^*, \beta_T^*)}{\partial \alpha} \hat{D}_{\alpha,\beta} + \frac{\partial \Gamma(\alpha_T^*, \beta_T^*)}{\partial \beta}, \quad \hat{D}_{\alpha,\beta} := - \left(\frac{\partial^2 \ell_T(\alpha_T^*, \beta_T^*)}{\partial \alpha \partial \alpha'} \right)^{-1} \frac{\partial^2 \ell_T(\alpha_T^*, \beta_T^*)}{\partial \alpha \partial \beta'}.$$

Unlike Jørgensen (2023), Alegre (2026) evaluates the Hessians at the estimated reference point η_T^* rather than at $(\hat{\alpha}(\beta_T^*), \beta_T^*)$. Under the same Bernstein–von Mises regularity conditions, together with Assumption 3 on the reference point, $K_{T,S}$ converges to $\hat{K}_S$ in probability, see Appendix A.3 for the full discussion.

5.2 Discussion of Partition Selection

The closest partition-selection method is Alegre (2026) that develops a Minimum Distance partition selection method based on minimizing worst-case bias. Alegre (2026) selects the admissible partition that minimizes $\hat{K}_S$. The main difference with the present paper is that they minimize worst-case bias for the object of interest, whereas we do so for the posterior mean of the object of interest. Their least-sensitive partition is

$$\hat{S}^* := \arg \min_{S \in \hat{S}_{AD}(\eta_T^*)} \hat{K}_S. \quad (16)$$

Under some regularity conditions discussed in appendix A.3, the robust posterior mean partition asymptotically agrees with the least-sensitive partition in Alegre (2026):

$$\Pr(S_T^* = \hat{S}^*) \longrightarrow 1 \quad \text{as } T \rightarrow \infty.$$

6. MONTE CARLO SIMULATION: AN AND SCHORFHEIDE 2007.

In this section, we apply the proposed method to find the least-sensitive partition within the New Keynesian model developed by [An and Schorfheide \(2007\)](#). This model has been widely utilized in the macroeconomic literature. We focus in the particular case of Bayesian estimation using the same prior distribution as in [An and Schorfheide \(2007\)](#). The log-linearized model is defined by:

$$\textbf{Euler Equation: } y_t = \mathbb{E}_t y_{t+1} + g_t - \mathbb{E}_t g_{t+1} - \frac{1}{\tau}(r_t - \mathbb{E}_t \pi_{t+1} - \mathbb{E}_t z_{t+1}), \quad (17)$$

$$\textbf{Phillips Curve: } \pi_t = \rho \mathbb{E}_t \pi_{t+1} + \kappa(y_t - g_t), \quad (18)$$

$$c_t = y_t - g_t, \quad (19)$$

$$\textbf{Monetary Policy: } r_t = \rho_r r_{t-1} + (1 - \rho_r)\psi_1 \pi_t + (1 - \rho_r)\psi_2(y_t - g_t) + \epsilon_{rt}, \quad (20)$$

$$g_t = \rho_g g_{t-1} + \epsilon_{gt}, \quad (21)$$

$$z_t = \rho_z z_{t-1} + \epsilon_{zt}. \quad (22)$$

where $\epsilon_{rt} \sim WN(0, \sigma_r^2)$, $\epsilon_{gt} \sim WN(0, \sigma_g^2)$, and $\epsilon_{zt} \sim WN(0, \sigma_z^2)$ are mutually uncorrelated structural shocks. The vector of parameters is:

$$\eta \equiv (\eta_1, \eta_2, \eta_3, \eta_4, \eta_5, \eta_6, \eta_7, \eta_8, \eta_9, \eta_{10}, \eta_{11}) \equiv (\rho, \tau, \kappa, \psi_1, \psi_2, \rho_r, \rho_g, \rho_z, \sigma_r^2, \sigma_g^2, \sigma_z^2).$$

For clarity, I use the notation of the right-hand side. The values of the Data Generating Process are the same as those used in [An and Schorfheide \(2007\)](#):

$$\eta_0 = (0.9975, 2, 0.33, 1.5, 0.125, 0.75, 0.95, 0.9, 0.04, 0.36, 0.09).$$

Suppose that the parameter of interest in this case is the reaction of the interest rate to changes in inflation, i.e, ψ_1 .

The identification problem in this particular example has been widely studied³ by [Komunjer and Ng \(2011\)](#), [Qu and Tkachenko \(2012, 2017\)](#) and [Kocięcki and](#)

³See [Iskrev \(2010\)](#), [Komunjer and Ng \(2011\)](#), [Qu and Tkachenko \(2012\)](#) for papers on local identification of DSGE. See [Qu and Tkachenko \(2017\)](#), [Al-Sadoon and Zwiernik \(2019\)](#) or [Kocięcki and Kolasa \(2023\)](#) on global identification.

Kolasa (2023)⁴. To obtain conditional identification we need to fix one parameter out of $\psi_2, \rho_r, \sigma_r^2$.

In this example, I will use quantiles of the prior distributions used in An and Schorfheide (2007) to obtain Σ_S . In particular, the $q_{\min} = 0.5\%$ and $q_{\max} = 99.5\%$ quantiles of the prior distribution for the parameters we fix.

TABLE 1. Normalization Matrix, Σ_S

Conservative Prior		
	$\eta_{j,\min}$	$\eta_{j,\max}$
ψ_2	0.084	1.370
ρ_r	0.067	0.932
σ_r^2	0.155	1.318

The first three rows reproduce a common practice in applied structural estimation: committing ex ante to a particular calibration–estimation partition. The results show that no fixed partition is uniformly robust across sample sizes. Fixing σ_r^2 is the most robust choice when $T = 150$, but becomes the least robust when $T = 1000$, whereas fixing ψ_2 becomes the most robust choice in the larger sample. Moreover, the cost of choosing the wrong partition increases sharply with the sample size. Under a 5% miscalibration, for example, the worst absolute relative effect rises from 3.06% when $T = 150$ to 18.46% when $T = 1000$. Thus, in this exercise, a larger sample makes the choice of partition more consequential rather than eliminating calibration risk.

The remaining rows show the gains from selecting the partition instead of imposing it in advance. The Oracle is an infeasible benchmark that selects the partition producing the smallest realized effect of miscalibration. Exact Bayes performs close to this best possible benchmark and substantially reduces the effects of miscalibration relative to the less robust fixed-partition choices. When $T = 150$,

⁴This paper has derived the analytical form for the identified set of this example.

the Asymptotic rule performs less well because it almost always selects the partition fixing ψ_2 , whereas Exact Bayes and the Oracle vary their selections more across samples.

When $T = 1000$, the three selection rules become much more similar. Exact Bayes, the Asymptotic rule, and the Oracle all select the partition fixing ψ_2 in nearly every sample and consequently produce similar absolute relative effects. The asymptotic approximation therefore performs well in the larger sample, whereas Exact Bayes provides a clear improvement in the smaller sample and remains close to the Oracle in both cases.

TABLE 2. Monte Carlo mean absolute relative effect of parameter miscalibration on γ_T (percent)

Partition Rule	$T = 150$			$T = 1000$		
	1%	3%	5%	1%	3%	5%
Fix ψ_2 always	0.61%	1.33%	2.29%	0.49%	1.48%	2.43%
Fix ρ_r always	0.72%	1.81%	3.06%	1.80%	6.86%	13.11%
Fix σ_r always	0.53%	0.91%	1.32%	4.38%	13.56%	18.46%
Asymptotic	0.61%	1.30%	2.21%	0.64%	1.93%	3.04%
Exact Bayes	0.40%	0.66%	1.02%	0.48%	1.50%	2.50%
Oracle	0.30%	0.61%	0.95%	0.48%	1.48%	2.43%

TABLE 3. Partition-selection frequencies

Rule	Miscalibration	$T = 150$					
		$T = 1000$					
		ρ_r	σ_r	ψ_2	ρ_r	σ_r	ψ_2
Asymptotic	–	8.0%	0.0%	92.0%	2.9%	2.9%	94.1%
Exact Bayes	–	16.0%	78.0%	6.0%	2.9%	0.0%	97.1%
Oracle	1%	38.0%	38.0%	24.0%	2.9%	0.0%	97.1%
Oracle	3%	26.0%	64.0%	10.0%	0.0%	0.0%	100.0%
Oracle	5%	28.0%	62.0%	10.0%	0.0%	0.0%	100.0%

The first three rows illustrate the prevailing practice in applied structural estimation: committing ex ante to a particular calibration–estimation partition rather than selecting it according to an explicit robustness criterion. Fixing σ_r^2 is the most robust fixed-partition benchmark when $T = 150$, but it becomes the least robust when $T = 1000$. Conversely, fixing ψ_2 is less robust than fixing σ_r^2 in the smaller sample, but in the larger sample it produces outcomes almost identical to those of the Oracle benchmark introduced below.

More generally, the dispersion in robustness across partitions increases sharply with the sample size. Under a 5% miscalibration, the mean absolute relative effect across the three fixed partitions ranges from 1.32% to 3.06% when $T = 150$, but from 2.43% to 18.46% when $T = 1000$. Hence, although Exact Bayes and the asymptotic approximation tend to select the same partition as the sample grows, choosing the wrong partition has substantially larger consequences in the larger sample.

The Oracle provides an infeasible benchmark: it selects the partition that would produce the smallest effect of miscalibration if the worst-case bias associated with each partition were known in advance. When $T = 150$, Exact Bayes performs close to this benchmark, whereas the Asymptotic rule produces noticeably larger changes in the posterior mean of ψ_1 . The selection frequencies help explain this difference. The Asymptotic rule almost always selects the partition fixing ψ_2 , while Exact Bayes and the Oracle select alternative partitions more frequently across simulated samples. Thus, in the smaller sample, the exact Bayesian criterion is better able to capture which partition is most robust in each case.

When $T = 1000$, the differences among the three rules become much smaller. Exact Bayes, the Asymptotic rule, and the Oracle all select the partition fixing ψ_2 in nearly every sample, and their resulting absolute relative effects are correspondingly similar. The asymptotic approximation therefore performs reasonably well in the larger sample, whereas the finite-sample Bayesian criterion provides a clear advantage when the sample is smaller.

7. SMETS AND WOUTERS 2007 APPLICATION

Smets and Wouters (2007) estimate a medium-scale closed-economy New Keynesian DSGE model using seven quarterly U.S. time series over the period 1966:1–2004:4. The model combines nominal price and wage rigidities with partial indexation, external habit formation, investment adjustment costs, variable capital utilization, and fixed costs in production. Its stochastic dynamics are driven by seven orthogonal structural shocks: productivity, risk-premium, investment-specific technology, exogenous-spending, monetary-policy, price-markup, and wage-markup shocks. A central use of the estimated model is to study the sources of fluctuations in output and inflation through forecast-error variance decompositions (FEVDs).

Smets and Wouters (2007) (hereafter SW07) find that the three shocks that primarily affect aggregate demand—the risk-premium, exogenous-spending, and investment-specific technology shocks—account for a large share of short-run output fluctuations. At medium and long horizons, output is instead driven mainly by productivity and wage-markup shocks. For inflation, price-markup shocks dominate at short horizons, whereas wage-markup shocks become relatively more important at longer horizons. We summarize these conclusions in a four-dimensional object. Let $V_x^{\mathcal{G}}(h; \eta)$ denote the share of the h -quarter forecast-error variance of variable x attributable to the shocks in group $\mathcal{G}$, and define

$$\mathcal{D} := \{\text{risk premium, exogenous spending, investment-specific technology}\},$$

$$\mathcal{S}_y := \{\text{productivity, wage markup}\}.$$

The object of interest is

$$\Gamma_{\text{FEVD}}(\eta) := \begin{pmatrix} V_y^{\mathcal{D}}(4; \eta) - V_y^{\mathcal{S}_y}(4; \eta) \\ V_y^{\mathcal{S}_y}(40; \eta) - V_y^{\mathcal{D}}(40; \eta) \\ V_{\pi}^p(4; \eta) - V_{\pi}^w(4; \eta) \\ V_{\pi}^w(40; \eta) - V_{\pi}^p(40; \eta) \end{pmatrix}, \quad (23)$$

where V_π^p and V_π^w denote the contributions of the price- and wage-markup shocks, respectively. The signs are chosen so that a positive value of each component agrees with the qualitative ordering emphasized by SW07.

The model contains 41 parameters. Table 4 below shows the list of estimated and calibrated parameters in SW07. The original partition estimate 36 and fix the remaining five according to

$$S_{\text{SW}}^c = \{\epsilon_w, \epsilon_p, \lambda_w, g_y, \delta\}, \quad (\epsilon_w, \epsilon_p, \lambda_w, g_y, \delta) = (10, 10, 1.5, 0.18, 0.025).$$

They argue that the depreciation rate δ and the exogenous-spending share g_y are difficult to estimate without including the corresponding investment and spending ratios in the measurement equations. They also treat the steady-state wage markup λ_w and the two Kimball-curvature parameters ϵ_w and ϵ_p as unidentified.

The subsequent analysis of Iskrev (2010, 2019) refines this identification argument. In particular, λ_w is locally identified, while the Calvo and Kimball-curvature parameters are not separately identified within the wage- and price-setting blocks:

$$(\xi_w, \epsilon_w) \quad \text{and} \quad (\xi_p, \epsilon_p)$$

each generate one local non-identification direction. Consequently, the likelihood cannot locally identify all 41 parameters simultaneously, but it can identify a 39-parameter block after fixing one parameter from each pair. Thus, only two fixed parameters are required for local identification; fixing λ_w , g_y , or δ is an additional calibration choice rather than an identification requirement.

TABLE 4. Common reference point for the Exact-Bayesian partition-selection exercise: structural parameters

Parameter	Definition	Reference Point η_T^*	Value source
φ	Investment-adjustment-cost parameter	5.74	Posterior mean
σ_c	Consumption curvature (inverse intertemporal elasticity)	1.38	Posterior mean
h	External habit formation	0.71	Posterior mean
ξ_w	Calvo wage-stickiness probability	0.70	Posterior mean
σ_l	Labor-supply curvature (inverse Frisch elasticity)	1.83	Posterior mean
ξ_p	Calvo price-stickiness probability	0.66	Posterior mean
ι_w	Wage indexation to past inflation	0.58	Posterior mean
ι_p	Price indexation to past inflation	0.24	Posterior mean
ψ	Capital-utilization-cost curvature	0.54	Posterior mean
Φ	Fixed-cost parameter (one plus the fixed-cost share)	1.60	Posterior mean
r_π	Monetary-policy response to inflation	2.04	Posterior mean
ρ	Interest-rate smoothing	0.81	Posterior mean
r_y	Monetary-policy response to the output gap	0.08	Posterior mean
$r_{\Delta y}$	Monetary-policy response to the change in the output gap	0.22	Posterior mean
$\bar{\pi}$	Quarterly steady-state inflation rate (percent)	0.78	Posterior mean
$100(\beta^{-1} - 1)$	Quarterly discount-rate transform (percent)	0.16	Posterior mean
$\bar{l}$	Steady-state labor / observation intercept	0.53	Posterior mean
$\bar{\gamma}$	Quarterly trend growth rate (percent)	0.43	Posterior mean
α	Capital share	0.19	Posterior mean
<i>Parameters calibrated in Smets and Wouters (2007)</i>			
ϵ_w	Kimball labor-market curvature	10.00	SW07 calibration
ϵ_p	Kimball goods-market curvature	10.00	SW07 calibration
λ_w	Steady-state labor-market markup	1.50	SW07 calibration
g_y	Exogenous-spending share of output	0.18	SW07 calibration
δ	Quarterly capital-depreciation rate	0.025	SW07 calibration

Table 5 presents the original priors used by SW07 to estimate the posterior distribution. We extend this list by specifying priors for the five parameters calibrated in SW07, so that, whenever a candidate partition treats any of these parameters as estimated, we can compute the Exact-Bayesian sensitivity using equation (8).

TABLE 5. Prior distributions used in the Exact-Bayesian partition-selection exercise: structural parameters

Parameter	Prior distribution		
	Distribution	Mean	St. dev.
φ	Normal	4.000	1.500
σ_c	Normal	1.500	0.375
h	Beta	0.700	0.100
ξ_w	Beta	0.500	0.100
σ_l	Normal	2.000	0.750
ξ_p	Beta	0.500	0.100
ι_w	Beta	0.500	0.150
ι_p	Beta	0.500	0.150
ψ	Beta	0.500	0.150
Φ	Normal	1.250	0.125
r_π	Normal	1.500	0.250
ρ	Beta	0.750	0.100
r_y	Normal	0.125	0.050
$r_{\Delta y}$	Normal	0.125	0.050
$\bar{\pi}$	Gamma	0.625	0.100
$100(\beta^{-1} - 1)$	Gamma	0.250	0.100
$\bar{l}$	Normal	0.000	2.000
$\bar{\gamma}$	Normal	0.400	0.100
α	Normal	0.300	0.050
ϵ_w	Gamma	10.000	5.000
ϵ_p	Gamma	10.000	5.000
λ_w	Shifted Gamma	1.250	0.150
g_y	Beta	0.180	0.025
δ	Beta	0.025	0.005

Based on Iskrev’s identification results, we construct the admissible set by requiring each partition to fix exactly one parameter from each of the two locally nonidentified pairs, (ξ_w, ϵ_w) and (ξ_p, ϵ_p) . This restriction yields four identification-compatible baseline combinations. We then allow each of the three additional parameters calibrated in SW07— λ_w , g_y , and δ —to be either fixed or estimated, since their calibration is not required for local identification. Combining the four baseline combinations with the $2^3 = 8$ possible choices for these three parameters produces a total of $4 \times 8 = 32$ admissible partitions. Thus, every candidate partition fixes between two and five parameters and is otherwise restricted to parameters calibrated in the original SW07 partition, except that ξ_w and ξ_p may be fixed in place of ϵ_w and ϵ_p , respectively, as implied by the identification trade-offs documented by Iskrev. For each parameter, we range-normalize using the 0.005 and 0.995 quantiles of its prior distribution.

TABLE 6. Normalization ranges for parameters eligible to be fixed

Parameter	Normalization range	
	$\eta_{j,min}$	$\eta_{j,max}$
ξ_w	0.252	0.748
ϵ_w	1.681	27.444
ξ_p	0.252	0.748
ϵ_p	1.681	27.444
λ_w	1.025	1.799
g_y	0.121	0.249
δ	0.014	0.040

Table 7 compares the sensitivity of the four-dimensional FEVD object across alternative calibration–estimation partitions. We report both the Exact-Bayesian sensitivity, computed using the finite-sample formula in equation (8), and its asymptotic-MLE counterpart implied by the Bernstein–von Mises approximation discussed in Section 5.1.

The main result is that the finite-sample and asymptotic sensitivities produce substantially different partition rankings. In particular, the partition selected by

Exact Bayes is ranked nineteenth by the asymptotic criterion, while the asymptotic winner is ranked twenty-fourth by Exact Bayes. In this application, using the exact Bayesian sensitivity can consequently lead to a markedly different choice of which structural parameters to fix and which to estimate.

TABLE 7. Exact-Bayesian and asymptotic-MLE partition sensitivities for the joint four-component Smets–Wouters FEVD target

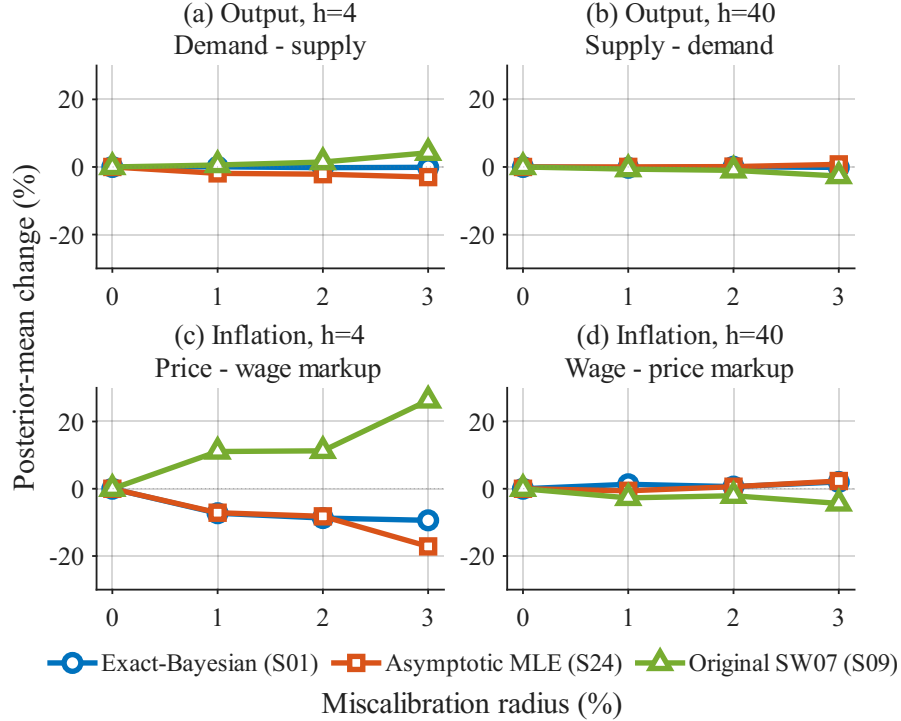
Fixed structural parameters	Bayes sensitivity $K_{\text{FEVD},S}^B$	MLE sensitivity $K_{\text{FEVD},S}^{\text{MLE}}$	Bayes rank	Assymp-MLE rank
$\epsilon_w, \epsilon_p, \lambda_w$	0.185	4.326	1	19
ϵ_w, ϵ_p	0.207	4.419	2	20
$\xi_w, \epsilon_p, g_y, \delta$	0.352	2.992	3	11
$\epsilon_w, \epsilon_p, \delta$	0.360	4.649	4	23
$\epsilon_w, \epsilon_p, g_y$	0.412	16.376	5	28
$\xi_w, \epsilon_p, \delta$	0.416	19.023	6	29
ξ_w, ϵ_p	0.479	51.524	7	32
ξ_w, ϵ_p, g_y	0.606	2.379	8	5
$\epsilon_w, \epsilon_p, \lambda_w, g_y, \delta$	0.646	6.417	9	26
ϵ_w, ξ_p, g_y	0.727	2.466	10	6
$\epsilon_w, \epsilon_p, \lambda_w, \delta$	0.779	4.613	11	21
$\xi_w, \epsilon_p, \lambda_w, \delta$	0.792	19.556	12	30
$\epsilon_w, \epsilon_p, \lambda_w, g_y$	0.838	5.709	13	25
ϵ_w, ξ_p	0.982	1.979	14	2
$\xi_w, \epsilon_p, \lambda_w, g_y$	0.986	2.923	15	9
$\epsilon_w, \epsilon_p, g_y, \delta$	1.012	14.060	16	27
$\xi_w, \xi_p, g_y, \delta$	1.136	3.308	17	14
ξ_w, ξ_p, λ_w	1.150	2.080	18	3
$\epsilon_w, \xi_p, \lambda_w, g_y, \delta$	1.273	4.120	19	18
$\epsilon_w, \xi_p, \lambda_w$	1.364	2.943	20	10
$\xi_w, \xi_p, \lambda_w, \delta$	1.376	2.575	21	7
$\xi_w, \xi_p, \lambda_w, g_y, \delta$	1.477	3.961	22	16
$\xi_w, \xi_p, \lambda_w, g_y$	1.504	5.383	23	24
ξ_w, ξ_p	1.508	1.599	24	1
ξ_w, ξ_p, δ	1.641	2.140	25	4
$\xi_w, \epsilon_p, \lambda_w, g_y, \delta$	1.650	4.095	26	17
$\epsilon_w, \xi_p, \delta$	1.663	2.654	27	8
$\epsilon_w, \xi_p, \lambda_w, g_y$	1.718	3.156	28	13
$\xi_w, \epsilon_p, \lambda_w$	1.822	35.049	29	31
$\epsilon_w, \xi_p, \lambda_w, \delta$	1.875	3.605	30	15
$\epsilon_w, \xi_p, g_y, \delta$	1.962	3.039	31	12
ξ_w, ξ_p, g_y	2.359	4.630	32	22

To assess the economic magnitude of these ranking differences, we re-estimate the model after perturbing the fixed parameters along the Exact-Bayesian worst-case miscalibration direction evaluated at the reference point:

$$\beta_{T,S}(r) = \beta_{T,S}^* + r \sqrt{|S^c|} \Sigma_S v_{1,S}^B, \quad r \in [0, 0.03],$$

where $v_{1,S}^B$ is the Exact-Bayesian worst-case direction for the aggregate FEVD vector. For each partition, we re-estimate the model over a grid of miscalibration sizes ranging from 0% to 3%. Because of computational constraints, we conduct this exercise for only three partitions: the original SW07 partition, the least-sensitive partition according to the Exact-Bayesian criterion, and the least-sensitive partition according to the asymptotic-MLE approximation. Figure 1 presents the results for the four components of the FEVD vector. For ease of comparison, the vertical axis reports the percentage change in each posterior mean relative to its corresponding value at $r = 0$.

FIGURE 1. Percentage changes in posterior means under miscalibration along the Exact-Bayesian worst-case direction



8. CONCLUSION

This paper develops a systematic method for deciding which structural parameters to estimate and which to calibrate in a Bayesian framework. Among a researcher-specified set of admissible partitions, the proposed criterion selects the partition that minimizes the worst-case local bias of the posterior mean of an object of interest induced by calibration errors. A central contribution is an exact finite-sample expression for the sensitivity of the posterior mean to the calibrated parameters. The expression can be computed from the restricted posterior distribution and does not rely on a Bernstein–von Mises approximation. Moreover, the procedure requires only conditional local identification of the parameters estimated under each candidate partition, rather than local identification of the full structural parameter vector.

Under suitable regularity conditions, the Exact-Bayesian criterion is asymptotically equivalent to its minimum-distance or MLE counterpart. The numerical results nevertheless show that this equivalence need not provide an accurate guide in empirically relevant finite samples. In the Monte Carlo exercise, the Exact-Bayesian rule remains close to the infeasible Oracle and substantially reduces the consequences of miscalibration relative to the asymptotic rule in the smaller sample. The two procedures become increasingly similar as the sample grows, consistent with the theoretical result. At the same time, the exercise shows that larger samples do not necessarily make the calibration choice irrelevant: the cost of imposing an unsuitable partition can instead become more pronounced.

The Smets–Wouters application reinforces the practical importance of using the finite-sample criterion. The Exact-Bayesian and asymptotic-MLE statistics generate markedly different rankings of the admissible partitions: the Exact-Bayesian winner is ranked nineteenth by the asymptotic criterion, while the asymptotic winner is ranked twenty-fourth by Exact Bayes. Re-estimating the model along the corresponding worst-case miscalibration directions also produces economically meaningful changes in the posterior FEVD summaries. Thus, replacing the exact posterior sensitivity with its asymptotic approximation can materially alter the recommendation about which parameters should be fixed.

More broadly, the results suggest that calibration should be treated as a source of specification risk rather than as an innocuous preliminary choice. The proposed procedure makes this risk explicit and provides a transparent way to combine the model-implied sensitivity of the target, identification restrictions, and external information about plausible calibration errors. The selected partition is deliberately local and object-specific, and therefore remains conditional on the reference point, the normalization ranges, and the admissible set specified by the researcher. These inputs should be reported clearly and, where relevant, examined through additional robustness exercises. The accompanying MATLAB/Dynare implementation makes such comparisons feasible for applied Bayesian DSGE analysis.

APPENDIX A: FURTHER DISCUSSIONS

A.1 *Unit-normalized sensitivities for reporting*

The partition-selection statistic normalizes perturbations in the fixed block using the matrix Σ_S . For transparency, it can also be useful to normalize the responses of the estimated parameters or of the object of interest. This subsection defines such transformations. They are reporting devices and do not alter the partition-selection criterion.

Let $A_T(S)$ and $G_T(S)$ be nonsingular diagonal matrices containing the scales in which changes in the estimated parameters and the object of interest are to be reported. Define

$$\mathcal{E}_{\alpha,\beta,T}(S) := A_T(S)^{-1} \frac{\partial \alpha_T(\beta; S)}{\partial \beta} \Sigma_S$$

and

$$\mathcal{E}_{\gamma,\beta,T}(S) := G_T(S)^{-1} \frac{\partial \gamma_T(\beta; S)}{\partial \beta} \Sigma_S.$$

When all components are nonzero, natural choices are

$$A_T(S) = \text{diag}(\alpha_T(\beta_T^*; S)), \quad G_T(S) = \text{diag}(\gamma_T(\beta_T^*; S)).$$

With these choices, the (j, i) element of $\mathcal{E}_{\alpha, \beta, T}(S)$ can be interpreted as a range-normalized elasticity. To first order, it gives the percentage change in the posterior mean of the j th estimated parameter induced by a change in the i th fixed parameter equal to one percent of its plausible range, holding the remaining fixed parameters constant. Analogously, the (j, i) element of $\mathcal{E}_{\gamma, \beta, T}(S)$ gives the percentage change in the j th component of the posterior mean of the object of interest induced by the same perturbation. The sign indicates the direction of the response.

Unlike a normalization based on unrestricted-posterior standard deviations, as in [Iskrev \(2019\)](#), this transformation does not require the full Fisher information matrix to be invertible. It uses only the restricted posterior sensitivity and the externally specified matrix Σ_S .

A.2 Reference Point Under Misspecification of the Restricted Model

Assumption 3 requires the reference point to converge to a point in the identified set N_0^{full} .

To obtain a reference point satisfying Assumption 3, consider the case where the researcher has a pseudo-reference point $\tilde{\eta}$, i.e., a reference point that may not satisfy the full score restrictions. Define $\eta^\dagger$ as the point in the full-score identified set closest to $\tilde{\eta}$:

$$\eta^\dagger \in \arg \min_{\eta \in N_0^{\text{full}}} \|\eta - \tilde{\eta}\|_2. \quad (24)$$

By construction, $\eta^\dagger \in N_0^{\text{full}}$. A sample analogue can be obtained by solving the following ridge-regularized maximum likelihood problem:

$$\eta_T^* \in \arg \max_{\eta \in \mathcal{N}} \left\{ \frac{1}{T} \ell_T(\eta) - \lambda_T \|\eta - \tilde{\eta}\|_2^2 \right\}. \quad (25)$$

The likelihood term pushes the estimator toward the full-score identified set, while the penalty selects the element of that set closest to the pseudo-reference point $\tilde{\eta}$. Under the likelihood analogues of the compactness, uniform-convergence, separation, and projection-uniqueness conditions in [Alegre \(2026, Appendix B.1\)](#), and provided that the penalty vanishes while dominating the stochastic

variation of the likelihood criterion along $\mathcal{N}_0$, we have

$$\eta_T^* \xrightarrow{p} \eta^\dagger.$$

Consequently, the resulting reference point satisfies Assumption 3, even when the restricted model used in the first-stage estimation is misspecified. The proof follows the same vanishing-penalty projection argument as Alegre (2026, Theorem 8); the only difference is that the quadratic full-score criterion used there is replaced by the average log likelihood.

A.3 Sensitivity relationships

For a fixed admissible partition S , let

$$\hat{\alpha}_T := \hat{\alpha}(\beta_T^*; S), \quad \bar{\alpha}_T := \mathbb{E}_T[\alpha], \quad V_T := \text{Var}_T(\alpha),$$

where $\mathbb{E}_T$ denotes expectation under $\pi_T(\alpha \mid Y_{1:T}, \beta_T^*; S)$. Define

$$s_{\beta,T}(a) := \nabla_{\beta} \ell_T(a, \beta_T^*; S)$$

and the first-order Taylor remainder

$$r_T(a) := s_{\beta,T}(a) - s_{\beta,T}(\hat{\alpha}_T) - \nabla_{\alpha'} s_{\beta,T}(\hat{\alpha}_T)(a - \hat{\alpha}_T).$$

ASSUMPTION 4 (Higher-order regularity). For every fixed admissible partition S , the following conditions hold.

- (i) *Likelihood regularity.* With probability approaching one, $(\hat{\alpha}_T, \beta_T^*)$ is an interior point and $\ell_T(\alpha, \beta; S)$ is twice continuously differentiable in a neighborhood of that point. There exists a fixed convex neighborhood $\mathcal{U}$ such that $\Pr(\hat{\alpha}_T \in \mathcal{U}) \rightarrow 1$, each component $s_{\beta_j,T}$ is twice continuously differentiable in α on $\mathcal{U}$, and

$$\max_{1 \leq j \leq d_{\beta}} \frac{1}{T} \sup_{a \in \mathcal{U}} \left\| \frac{\partial^2 s_{\beta_j,T}(a)}{\partial a \partial a'} \right\|_2 = O_p(1).$$

Moreover, $\hat{I}_{\alpha,\alpha}$ is nonsingular with probability approaching one and

$$\left\| \hat{I}_{\alpha,\alpha}^{-1} \right\|_2 + \left\| \hat{I}_{\alpha,\beta} \right\|_2 = O_p(1).$$

(ii) *Posterior moments.* The posterior variance and third centered moment satisfy

$$TV_T = \widehat{I}_{\alpha,\alpha}^{-1} + o_p(1), \quad \mathbb{E}_T \left[\left\| \sqrt{T}(\alpha - \widehat{\alpha}_T) \right\|_2^3 \right] = O_p(1).$$

(iii) *Posterior-tail control.* The contribution of the Taylor remainder outside $\mathcal{U}$ is asymptotically negligible:

$$\mathbb{E}_T [\|\alpha - \bar{\alpha}_T\|_2 \|r_T(\alpha)\|_2 \mathbf{1}\{\alpha \notin \mathcal{U}\}] = o_p(1).$$

(iv) *Object and reference-point regularity.* The function $\Gamma(\alpha, \beta; S)$ is twice continuously differentiable in α , and $\Gamma_\beta(\alpha, \beta; S)$ is continuously differentiable in α , on a neighborhood of (α^*, β^*) . The relevant derivatives are bounded on this neighborhood.

Define the first-order remainder

$$r_{\Gamma,T}(a) := \Gamma(a, \beta_T^*; S) - \Gamma(\widehat{\alpha}_T, \beta_T^*; S) - \Gamma_\alpha(\widehat{\alpha}_T, \beta_T^*; S)(a - \widehat{\alpha}_T).$$

The posterior satisfies

$$E_T [\|\Gamma_\beta(\alpha, \beta_T^*; S) - \Gamma_\beta(\widehat{\alpha}_T, \beta_T^*; S)\|_2] = o_p(1),$$

and

$$E_T [\|r_{\Gamma,T}(\alpha)\|_2 \|s_{\beta,T}(\alpha) - E_T[s_{\beta,T}(\alpha)]\|_2] = o_p(1).$$

Moreover, the Hessian blocks are asymptotically unchanged when moving from the conditional MLE to the reference point:

$$\left\| \widehat{I}_{\alpha,\alpha} - \widehat{I}_{\alpha,\alpha}^* \right\|_2 + \left\| \widehat{I}_{\alpha,\beta} - \widehat{I}_{\alpha,\beta}^* \right\|_2 = o_p(1).$$

The first condition controls the local quadratic remainder in the β -score. The second strengthens the usual Bernstein–von Mises approximation to the posterior moments used in the proof, while the third controls posterior draws outside the local likelihood neighborhood.

THEOREM 3 (Asymptotic equivalence of Bayesian and MLE sensitivity). *Fix an admissible partition S . Suppose that conditions underlying equation (7) and Assumption 4 hold. Then,*

$$\left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} = \widehat{I}_{\alpha, \alpha}^{-1} \widehat{I}_{\alpha, \beta} + o_p(1),$$

PROOF. See Appendix B. □

LEMMA 4. *Fix an admissible partition S . Suppose that the conditions underlying equation (8), Assumption 3, and Assumption 4 hold. Then,*

$$\left. \frac{\partial \gamma_T(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} = \widehat{D}_{\gamma, \beta} + o_p(1).$$

Consequently,

$$K_{T,S} - \widehat{K}_S = o_p(1).$$

PROOF. See Appendix B. □

ASSUMPTION 5. The population admissible set $\mathcal{S}_{\text{AD}}^0$ is nonempty, and

$$\Pr\left(\widehat{\mathcal{S}}_{\text{AD},T} = \mathcal{S}_{\text{AD},T} = \mathcal{S}_{\text{AD}}^0\right) \rightarrow 1.$$

Moreover, for every $S \in \mathcal{S}_{\text{AD}}^0$,

$$K_S := \text{plim}_{T \rightarrow \infty} \widehat{K}_S$$

exists, and the population problem

$$S^* := \arg \min_{S \in \mathcal{S}_{\text{AD}}^0} K_S$$

has a unique solution.

ASSUMPTION 6. The problem $\min_{S \in \mathcal{S}_{\text{AD}}} K_S$ has a unique solution, where $K_S := \text{plim} \widehat{K}_S$.

THEOREM 5. Suppose that Assumptions 2–6 hold for every $S \in \mathcal{S}_{AD}$. Then,

$$\Pr\left(S_T^* = \widehat{S}^*\right) \longrightarrow 1 \quad \text{as } T \rightarrow \infty.$$

PROOF. See Appendix B. □

APPENDIX B: MAIN PROOFS

PROOF. **Showing** $\partial \alpha_T(\beta; S) / \partial \beta = \text{Cov}_{\pi_T}(\alpha, \nabla_\beta \ell_T)$:

Let

$$f(\alpha | y_{1:t}, \beta) = \frac{f(y_{1:t} | \alpha, \beta) \pi(\alpha)}{\int f(y_{1:t} | \alpha, \beta) \pi(\alpha) d\alpha} := \frac{N(\alpha, \beta)}{D(\beta)}.$$

Then,

$$\frac{\partial N(\alpha, \beta)}{\partial \beta} = \frac{\partial f(y_{1:t} | \alpha, \beta)}{\partial \beta} \pi(\alpha) = \frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} f(y_{1:t} | \alpha, \beta) \pi(\alpha),$$

and

$$\frac{\partial D(\beta)}{\partial \beta} = \int \frac{\partial f(y_{1:t} | \alpha, \beta)}{\partial \beta} \pi(\alpha) d\alpha = \int \frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} f(y_{1:t} | \alpha, \beta) \pi(\alpha) d\alpha.$$

Notice that,

$$\frac{\partial N(\alpha, \beta)}{\partial \beta} \cdot \frac{1}{D(\beta)} = \frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} \frac{f(y_{1:t} | \alpha, \beta) \pi(\alpha)}{\int f(y_{1:t} | \alpha, \beta) \pi(\alpha) d(\alpha)} = \frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} \cdot f(\alpha | y_{1:t}, \beta),$$

and

$$\begin{aligned} \frac{\partial D(\beta)}{\partial \beta} \cdot \frac{N(\alpha, \beta)}{D(\beta)^2} &= \int \frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} \frac{f(y_{1:t} | \alpha, \beta) \pi(\alpha)}{\int f(y_{1:t} | \alpha, \beta) \pi(\alpha) d\alpha} d\alpha \cdot \frac{f(y_{1:t} | \alpha, \beta) \pi(\alpha)}{\int f(y_{1:t} | \alpha, \beta) \pi(\alpha) d\alpha} \\ &= \int \frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} f(\alpha | y_{1:t}, \beta) d\alpha f(\alpha | y_{1:t}, \beta) \end{aligned}$$

Therefore,

$$\begin{aligned}
\frac{\partial}{\partial \beta} f(\alpha | y_{1:t}, \beta) &= \frac{\partial N(\alpha, \beta)}{\partial \beta} \cdot \frac{1}{D(\beta)} - \frac{\partial D(\beta)}{\partial \beta} N(\alpha, \beta) \cdot \frac{1}{D(\beta)^2} \\
&= \left[\frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} - \int \frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} f(\alpha | y_{1:t}, \beta) d\alpha \right] f(\alpha | y_{1:t}, \beta) \\
&= \left[\frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} - E_\alpha \left(\frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} \right) \right] f(\alpha | y_{1:t}, \beta)
\end{aligned}$$

Using this expression to $\partial \alpha_t(\beta) / \partial \beta$:

$$\begin{aligned}
\frac{\partial \alpha_t(\beta; S)}{\partial \beta} &= \int \alpha \left[\frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} - E_\alpha \left(\frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} \right) \right] f(\alpha | y_{1:t}, \beta) d\alpha \\
&= \text{Cov}_\alpha \left(\alpha, \frac{\partial \log f(y_{1:t} | \alpha, \beta)}{\partial \beta} \right),
\end{aligned}$$

□

PROOF OF LEMMA 1. Fix a partition $S \in \mathcal{S}_{\text{AD}}(\eta_T^*)$. All statements below are conditional on $Y_{1:T}$ and on the realized reference point η_T^* .

Let $m_S := |S^c|$. If $m_S = 0$, there are no fixed parameters, $K_{T,S} = 0$, and

$$R_T(S; \eta_T^*, \Sigma_S, \epsilon) = 0,$$

so the result holds trivially. Assume hereafter that $m_S > 0$.

Write

$$D_{T,S} := \frac{\partial \gamma_T(\beta_T^*; S)}{\partial \beta'}$$

and define the first-order remainder

$$\mathcal{R}_{T,S}(\beta) := \gamma_T(\beta; S) - \gamma_T(\beta_T^*; S) - D_{T,S}(\beta - \beta_T^*).$$

Since $\gamma_T(\cdot; S)$ is Fréchet differentiable at β_T^* ,

$$\omega_{T,S}(\delta) := \sup_{\substack{0 < \|\beta - \beta_T^*\|_2 \leq \delta \\ \beta \in B_S}} \frac{\|\mathcal{R}_{T,S}(\beta)\|_2}{\|\beta - \beta_T^*\|_2} \longrightarrow 0 \quad \text{as } \delta \downarrow 0.$$

For every $\beta_0 \in B_\epsilon(\beta_T^*; S)$,

$$\|\beta_0 - \beta_T^*\|_2 = \|\Sigma_S \Sigma_S^{-1}(\beta_0 - \beta_T^*)\|_2 \leq \|\Sigma_S\|_2 \epsilon \sqrt{m_S}.$$

Consequently,

$$\begin{aligned} \sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|\mathcal{R}_{T,S}(\beta_0)\|_2 &\leq \epsilon \sqrt{m_S} \|\Sigma_S\|_2 \omega_{T,S}(\epsilon \sqrt{m_S} \|\Sigma_S\|_2) \\ &= o(\epsilon). \end{aligned}$$

Thus, uniformly over $\beta_0 \in B_\epsilon(\beta_T^*; S)$,

$$\gamma_T(\beta_0; S) - \gamma_T(\beta_T^*; S) = D_{T,S}(\beta_0 - \beta_T^*) + \mathcal{R}_{T,S}(\beta_0),$$

or, equivalently,

$$\gamma_T(\beta_T^*; S) - \gamma_T(\beta_0; S) = D_{T,S}(\beta_T^* - \beta_0) - \mathcal{R}_{T,S}(\beta_0).$$

By the triangle inequality, for every $\beta_0 \in B_\epsilon(\beta_T^*; S)$,

$$\|\gamma_T(\beta_T^*; S) - \gamma_T(\beta_0; S)\|_2 \leq \|D_{T,S}(\beta_T^* - \beta_0)\|_2 + \|\mathcal{R}_{T,S}(\beta_0)\|_2.$$

Taking suprema gives

$$\begin{aligned} &\sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|\gamma_T(\beta_T^*; S) - \gamma_T(\beta_0; S)\|_2 \\ &\leq \sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|D_{T,S}(\beta_T^* - \beta_0)\|_2 + \sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|\mathcal{R}_{T,S}(\beta_0)\|_2. \end{aligned} \quad (26)$$

Conversely,

$$D_{T,S}(\beta_T^* - \beta_0) = \gamma_T(\beta_T^*; S) - \gamma_T(\beta_0; S) + \mathcal{R}_{T,S}(\beta_0),$$

and hence

$$\|D_{T,S}(\beta_T^* - \beta_0)\|_2 \leq \|\gamma_T(\beta_T^*; S) - \gamma_T(\beta_0; S)\|_2 + \|\mathcal{R}_{T,S}(\beta_0)\|_2.$$

Taking suprema gives

$$\sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|D_{T,S}(\beta_T^* - \beta_0)\|_2$$

$$\leq \sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|\gamma_T(\beta_T^*; S) - \gamma_T(\beta_0; S)\|_2 + \sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|\mathcal{R}_{T,S}(\beta_0)\|_2. \quad (27)$$

Combining (26) and (27),

$$\left| \sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|\gamma_T(\beta_T^*; S) - \gamma_T(\beta_0; S)\|_2 - \sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|D_{T,S}(\beta_T^* - \beta_0)\|_2 \right| \leq \sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|\mathcal{R}_{T,S}(\beta_0)\|_2 = o(\epsilon). \quad (28)$$

Therefore,

$$R_T(S; \eta_T^*, \Sigma_S, \epsilon) = \sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|D_{T,S}(\beta_T^* - \beta_0)\|_2 + o(\epsilon). \quad (29)$$

Because $\beta_T^* \in \text{int}(B_S)$, for all sufficiently small $\epsilon > 0$ the ellipsoid defining $B_\epsilon(\beta_T^*; S)$ is contained in B_S . Hence

$$B_\epsilon(\beta_T^*; S) = \{\beta_T^* - \epsilon\sqrt{m_S}\Sigma_S x : \|x\|_2 \leq 1\}.$$

Using the change of variables

$$x := \frac{1}{\epsilon\sqrt{m_S}} \Sigma_S^{-1}(\beta_T^* - \beta_0),$$

we obtain

$$\begin{aligned} \sup_{\beta_0 \in B_\epsilon(\beta_T^*; S)} \|D_{T,S}(\beta_T^* - \beta_0)\|_2 &= \epsilon\sqrt{m_S} \sup_{\|x\|_2 \leq 1} \|D_{T,S}\Sigma_S x\|_2 \\ &= \epsilon\sqrt{m_S} \|D_{T,S}\Sigma_S\|_2, \end{aligned}$$

where the last equality follows from the definition of the induced Euclidean operator norm.

Recalling that

$$K_{T,S} := \sqrt{|S^c|} \left\| \frac{\partial \gamma_T(\beta_T^*; S)}{\partial \beta'} \Sigma_S \right\|_2 = \sqrt{m_S} \|D_{T,S}\Sigma_S\|_2,$$

equation (29) yields

$$R_T(S; \eta_T^*, \Sigma_S, \epsilon) = \epsilon K_{T,S} + o(\epsilon).$$

This completes the proof. □

PROOF OF LEMMA 2. All statements below are conditional on $Y_{1:T}$ and on the realized reference point η_T^* . Let

$$\mathcal{S}_{T,0} := \mathcal{S}_{\text{AD}}(\eta_T^*), \quad W_{T,S}(\epsilon) := R_T(S; \eta_T^*, \Sigma_S, \epsilon).$$

Since $d_\eta < \infty$, the number of possible partitions is finite, and hence $\mathcal{S}_{T,0}$ is finite. Let S_T^* denote the unique element of

$$\arg \min_{S \in \mathcal{S}_{T,0}} K_{T,S}.$$

The uniqueness of S_T^* follows from Assumption 6.

By Lemma 1, for every $S \in \mathcal{S}_{T,0}$,

$$W_{T,S}(\epsilon) = \epsilon K_{T,S} + o_{T,S}(\epsilon) \quad \text{as } \epsilon \downarrow 0.$$

Equivalently, for $0 < \epsilon$ sufficiently small,

$$W_{T,S}(\epsilon) = \epsilon \{K_{T,S} + r_{T,S}(\epsilon)\}, \quad r_{T,S}(\epsilon) \longrightarrow 0 \quad \text{as } \epsilon \downarrow 0.$$

Because $\mathcal{S}_{T,0}$ is finite,

$$\rho_T(\epsilon) := \max_{S \in \mathcal{S}_{T,0}} |r_{T,S}(\epsilon)| \longrightarrow 0 \quad \text{as } \epsilon \downarrow 0.$$

If $\mathcal{S}_{T,0} = \{S_T^*\}$, the result is immediate. Otherwise, define the strict sensitivity gap

$$\Delta_T := \min_{\substack{S \in \mathcal{S}_{T,0} \\ S \neq S_T^*}} \left\{ K_{T,S} - K_{T,S_T^*} \right\}.$$

By the uniqueness of S_T^* ,

$$\Delta_T > 0.$$

Since $\rho_T(\epsilon) \rightarrow 0$, there exists $E_T > 0$ such that, for every $0 < \epsilon \leq E_T$,

$$\rho_T(\epsilon) < \frac{\Delta_T}{3}.$$

Hence, for any $S \in \mathcal{S}_{T,0}$ with $S \neq S_T^*$, and any $0 < \epsilon \leq E_T$,

$$\begin{aligned}
W_{T,S}(\epsilon) - W_{T,S_T^*}(\epsilon) &= \epsilon \left\{ K_{T,S} - K_{T,S_T^*} \right\} + \epsilon \left\{ r_{T,S}(\epsilon) - r_{T,S_T^*}(\epsilon) \right\} \\
&\geq \epsilon \Delta_T - 2\epsilon \rho_T(\epsilon) \\
&> \epsilon \Delta_T - \frac{2}{3} \epsilon \Delta_T \\
&= \frac{1}{3} \epsilon \Delta_T > 0.
\end{aligned}$$

Therefore,

$$W_{T,S_T^*}(\epsilon) < W_{T,S}(\epsilon)$$

for every $S \neq S_T^*$ and every $0 < \epsilon \leq E_T$. Thus, S_T^* is the unique minimizer of the robust posterior risk over $\mathcal{S}_{T,0}$ for all such ϵ . Since S_T^* is also the unique minimizer of $K_{T,S}$, we obtain

$$\arg \min_{S \in \mathcal{S}_{\text{AD}}(\eta_T^*)} R_T(S; \eta_T^*, \Sigma_S, \epsilon) = \arg \min_{S \in \mathcal{S}_{\text{AD}}(\eta_T^*)} K_{T,S} = \{S_T^*\}, \quad 0 < \epsilon \leq E_T.$$

This proves the lemma. □

PROOF. (Proof of Theorem 3). Fix an admissible partition S and suppress its dependence from the notation. Let

$$\hat{\alpha} := \hat{\alpha}(\beta_T^*; S)$$

denote the conditional maximum likelihood estimator when the calibrated block is fixed at β_T^* .

All posterior expectations, variances, and covariances below are conditional on $Y_{1:T}$ and on the realized first-stage calibration β_T^* . To simplify notation, write

$$\mathbb{E}_T[h(\alpha)] := \int h(\alpha) \pi_T(\alpha \mid Y_{1:T}, \beta_T^*; S) d\alpha,$$

and define Var_T and Cov_T analogously. In particular, let

$$V_T := \text{Var}_T(\alpha).$$

We prove the result one calibrated parameter at a time. Fix $j \in \{1, \dots, d_\beta\}$ and define the scalar score associated with the j th calibrated parameter by

$$s_{j,T}(\alpha, \beta) := \frac{\partial \ell_T(\alpha, \beta; S)}{\partial \beta_j}.$$

By equation (7), the j th column of the exact Bayesian sensitivity matrix is

$$\left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta_j} \right|_{\beta=\beta_T^*} = \text{Cov}_T(\alpha, s_{j,T}(\alpha, \beta_T^*)) . \quad (30)$$

Define

$$x := \alpha - \hat{\alpha}, \quad \bar{x}_T := \mathbb{E}_T[x],$$

and let

$$g_{j,T} := \left. \frac{\partial s_{j,T}(a, \beta_T^*)}{\partial a} \right|_{a=\hat{\alpha}} = \frac{\partial^2 \ell_T(\hat{\alpha}, \beta_T^*; S)}{\partial \alpha \partial \beta_j}.$$

Thus, $g_{j,T}$ is a d_α -dimensional column vector.

For each α , define the exact Taylor remainder

$$r_{j,T}(\alpha) := s_{j,T}(\alpha, \beta_T^*) - s_{j,T}(\hat{\alpha}, \beta_T^*) - g'_{j,T}(\alpha - \hat{\alpha}).$$

It follows that

$$s_{j,T}(\alpha, \beta_T^*) = s_{j,T}(\hat{\alpha}, \beta_T^*) + g'_{j,T}x + r_{j,T}(\alpha). \quad (31)$$

Substituting equation (31) into equation (30) gives

$$\begin{aligned} \left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta_j} \right|_{\beta=\beta_T^*} &= \text{Cov}_T(\alpha, s_{j,T}(\hat{\alpha}, \beta_T^*)) \\ &\quad + \text{Cov}_T(\alpha, g'_{j,T}x) + \text{Cov}_T(\alpha, r_{j,T}(\alpha)). \end{aligned} \quad (32)$$

The first term on the right-hand side is zero because $s_{j,T}(\hat{\alpha}, \beta_T^*)$ is constant conditional on the data. Notice that this score component need not itself be zero: $\hat{\alpha}$ sets the α -score equal to zero, not the β_j -score.

For the second term, since $x = \alpha - \hat{\alpha}$ and $\hat{\alpha}$ is fixed conditional on the data,

$$\text{Cov}_T(\alpha, g'_{j,T}x) = \mathbb{E}_T[(x - \bar{x}_T)(x - \bar{x}_T)'] g_{j,T}$$

$$= V_T g_{j,T}. \quad (33)$$

Consequently,

$$\left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta_j} \right|_{\beta=\beta_T^*} = V_T g_{j,T} + R_{j,T}, \quad (34)$$

where

$$R_{j,T} := \text{Cov}_T(\alpha, r_{j,T}(\alpha)).$$

We next show that $R_{j,T} = o_p(1)$. Let U be the convex posterior-concentration neighborhood appearing in the likelihood regularity condition. Since $\hat{\alpha} \xrightarrow{p} \alpha^*$, we may work on the event $\hat{\alpha} \in U$, whose probability approaches one.

For every posterior draw $\alpha \in U$, the line segment between $\hat{\alpha}$ and α is contained in U . Taylor's theorem with integral remainder therefore gives

$$r_{j,T}(\alpha) = \int_0^1 (1-t) x' \frac{\partial^2 s_{j,T}(\hat{\alpha} + tx, \beta_T^*)}{\partial \alpha \partial \alpha'} x dt. \quad (35)$$

Define

$$M_{j,T} := \frac{1}{T} \sup_{a \in U} \left\| \frac{\partial^2 s_{j,T}(a, \beta_T^*)}{\partial a \partial a'} \right\|_2.$$

By the likelihood regularity condition,

$$M_{j,T} = O_p(1).$$

Using equation (35), for $\alpha \in U$ we have

$$\begin{aligned} |r_{j,T}(\alpha)| &\leq \int_0^1 (1-t) \left\| \frac{\partial^2 s_{j,T}(\hat{\alpha} + tx, \beta_T^*)}{\partial \alpha \partial \alpha'} \right\|_2 \|x\|_2^2 dt \\ &\leq \frac{T}{2} M_{j,T} \|x\|_2^2. \end{aligned} \quad (36)$$

Since $\mathbb{E}_T[x - \bar{x}_T] = 0$, the covariance remainder can be written as

$$R_{j,T} = \mathbb{E}_T[(x - \bar{x}_T) r_{j,T}(\alpha)].$$

Decomposing this expectation over U and U^c ,

$$\begin{aligned} R_{j,T} &= \mathbb{E}_T \left[(x - \bar{x}_T) r_{j,T}(\alpha) \mathbf{1}\{\alpha \in U\} \right] \\ &\quad + \mathbb{E}_T \left[(x - \bar{x}_T) r_{j,T}(\alpha) \mathbf{1}\{\alpha \notin U\} \right]. \end{aligned} \quad (37)$$

The second term in equation (37) is $o_p(1)$ by the posterior-tail condition. For the first term, equation (36) gives

$$\begin{aligned} \|R_{j,T}\|_2 &\leq \frac{T}{2} M_{j,T} \mathbb{E}_T \left[\|x - \bar{x}_T\|_2 \|x\|_2^2 \right] + o_p(1) \\ &\leq \frac{T}{2} M_{j,T} \left\{ \mathbb{E}_T \|x\|_2^3 + \|\bar{x}_T\|_2 \mathbb{E}_T \|x\|_2^2 \right\} + o_p(1). \end{aligned} \quad (38)$$

By Jensen's inequality,

$$\|\bar{x}_T\|_2 = \|\mathbb{E}_T[x]\|_2 \leq \mathbb{E}_T \|x\|_2.$$

Moreover, Hölder's inequality implies

$$\mathbb{E}_T \|x\|_2 \leq \left(\mathbb{E}_T \|x\|_2^3 \right)^{1/3}$$

and

$$\mathbb{E}_T \|x\|_2^2 \leq \left(\mathbb{E}_T \|x\|_2^3 \right)^{2/3}.$$

It follows that

$$\|\bar{x}_T\|_2 \mathbb{E}_T \|x\|_2^2 \leq \mathbb{E}_T \|x\|_2^3.$$

Substituting this inequality into equation (38) yields

$$\|R_{j,T}\|_2 \leq T M_{j,T} \mathbb{E}_T \left[\|\alpha - \hat{\alpha}\|_2^3 \right] + o_p(1). \quad (39)$$

By the posterior moment condition,

$$T \mathbb{E}_T \left[\|\alpha - \hat{\alpha}\|_2^3 \right] = o_p(1),$$

and $M_{j,T} = O_p(1)$. Therefore,

$$R_{j,T} = o_p(1). \quad (40)$$

Returning to equation (34), define

$$\hat{I}_{\alpha, \beta_j} := \frac{1}{T} \frac{\partial^2 \ell_T(\hat{\alpha}, \beta_T^*; S)}{\partial \alpha \partial \beta_j} = \frac{1}{T} g_{j, T}.$$

Equations (34) and (40) imply

$$\left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta_j} \right|_{\beta = \beta_T^*} = (TV_T) \hat{I}_{\alpha, \beta_j} + o_p(1). \quad (41)$$

By the posterior variance approximation,

$$TV_T = \hat{I}_{\alpha, \alpha}^{-1} + o_p(1),$$

where

$$\hat{I}_{\alpha, \alpha} := -\frac{1}{T} \frac{\partial^2 \ell_T(\hat{\alpha}, \beta_T^*; S)}{\partial \alpha \partial \alpha'}.$$

Since $\hat{I}_{\alpha, \beta_j} = O_p(1)$, equation (41) gives

$$\left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta_j} \right|_{\beta = \beta_T^*} = \hat{I}_{\alpha, \alpha}^{-1} \hat{I}_{\alpha, \beta_j} + o_p(1). \quad (42)$$

It remains to identify the leading term with the sensitivity of the conditional maximum likelihood estimator. The conditional MLE satisfies

$$\frac{\partial \ell_T(\hat{\alpha}(\beta; S), \beta; S)}{\partial \alpha} = 0. \quad (43)$$

Differentiating equation (43) with respect to the scalar parameter β_j and evaluating at $\beta = \beta_T^*$ gives

$$\begin{aligned} & \left. \frac{\partial^2 \ell_T(\hat{\alpha}, \beta_T^*; S)}{\partial \alpha \partial \alpha'} \frac{\partial \hat{\alpha}(\beta; S)}{\partial \beta_j} \right|_{\beta = \beta_T^*} \\ & + \frac{\partial^2 \ell_T(\hat{\alpha}, \beta_T^*; S)}{\partial \alpha \partial \beta_j} = 0. \end{aligned} \quad (44)$$

Since the Hessian with respect to α is nonsingular with probability approaching one, the implicit function theorem gives

$$\left. \frac{\partial \hat{\alpha}(\beta; S)}{\partial \beta_j} \right|_{\beta = \beta_T^*} = - \left[\frac{\partial^2 \ell_T(\hat{\alpha}, \beta_T^*; S)}{\partial \alpha \partial \alpha'} \right]^{-1} \frac{\partial^2 \ell_T(\hat{\alpha}, \beta_T^*; S)}{\partial \alpha \partial \beta_j}$$

$$= \widehat{I}_{\alpha, \alpha}^{-1} \widehat{I}_{\alpha, \beta_j}. \quad (45)$$

Combining equations (42) and (45), we obtain

$$\left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta_j} \right|_{\beta=\beta_T^*} - \left. \frac{\partial \widehat{\alpha}(\beta; S)}{\partial \beta_j} \right|_{\beta=\beta_T^*} = o_p(1). \quad (46)$$

Because j was arbitrary, equation (46) holds for every $j = 1, \dots, d_\beta$. Let

$$\Delta_{j,T} := \left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta_j} \right|_{\beta=\beta_T^*} - \left. \frac{\partial \widehat{\alpha}(\beta; S)}{\partial \beta_j} \right|_{\beta=\beta_T^*}.$$

The difference between the two full sensitivity matrices is

$$\left(\Delta_{1,T} \cdots \Delta_{d_\beta,T} \right).$$

Since d_β is fixed,

$$\begin{aligned} & \left\| \left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} - \left. \frac{\partial \widehat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} \right\|_F \\ & \leq \sqrt{d_\beta} \max_{1 \leq j \leq d_\beta} \|\Delta_{j,T}\|_2 = o_p(1). \end{aligned}$$

Therefore,

$$\left. \frac{\partial \alpha_T(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} - \left. \frac{\partial \widehat{\alpha}(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} = o_p(1),$$

which proves the result. $\square$

PROOF. (Proof of Lemma 4). Fix an admissible partition S and suppress its dependence from the notation. All posterior expectations and covariances below are conditional on $Y_{1:T}$ and on β_T^* .

By equation (8),

$$\begin{aligned} \left. \frac{\partial \gamma_T(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} &= E_T[\Gamma_\beta(\alpha, \beta_T^*; S)] \\ &\quad + \text{Cov}_T(\Gamma(\alpha, \beta_T^*; S), s_{\beta,T}(\alpha)). \end{aligned} \quad (47)$$

By the object-regularity condition,

$$E_T[\Gamma_\beta(\alpha, \beta_T^*; S)] = \Gamma_\beta(\hat{\alpha}_T, \beta_T^*; S) + o_p(1).$$

Next, by the definition of $r_{\Gamma, T}$,

$$\Gamma(\alpha, \beta_T^*; S) = \Gamma(\hat{\alpha}_T, \beta_T^*; S) + \Gamma_\alpha(\hat{\alpha}_T, \beta_T^*; S)(\alpha - \hat{\alpha}_T) + r_{\Gamma, T}(\alpha).$$

Since the first term is constant conditional on the data,

$$\begin{aligned} \text{Cov}_T(\Gamma(\alpha, \beta_T^*; S), s_{\beta, T}(\alpha)) \\ = \Gamma_\alpha(\hat{\alpha}_T, \beta_T^*; S) \text{Cov}_T(\alpha, s_{\beta, T}(\alpha)) + \text{Cov}_T(r_{\Gamma, T}(\alpha), s_{\beta, T}(\alpha)). \end{aligned}$$

Moreover,

$$\begin{aligned} \|\text{Cov}_T(r_{\Gamma, T}(\alpha), s_{\beta, T}(\alpha))\|_2 &\leq E_T[\|r_{\Gamma, T}(\alpha)\|_2 \|s_{\beta, T}(\alpha) - E_T[s_{\beta, T}(\alpha)]\|_2] \\ &= o_p(1). \end{aligned}$$

Therefore,

$$\text{Cov}_T(\Gamma(\alpha, \beta_T^*; S), s_{\beta, T}(\alpha)) = \Gamma_\alpha(\hat{\alpha}_T, \beta_T^*; S) \text{Cov}_T(\alpha, s_{\beta, T}(\alpha)) + o_p(1).$$

By equation (7) and Theorem 3,

$$\text{Cov}_T(\alpha, s_{\beta, T}(\alpha)) = \hat{I}_{\alpha, \alpha}^{-1} \hat{I}_{\alpha, \beta} + o_p(1).$$

Since $\Gamma_\alpha(\hat{\alpha}_T, \beta_T^*; S) = O_p(1)$, equation (47) consequently gives

$$\begin{aligned} \left. \frac{\partial \gamma_T(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} &= \Gamma_\beta(\hat{\alpha}_T, \beta_T^*; S) \\ &\quad + \Gamma_\alpha(\hat{\alpha}_T, \beta_T^*; S) \hat{I}_{\alpha, \alpha}^{-1} \hat{I}_{\alpha, \beta} + o_p(1). \end{aligned} \tag{48}$$

It remains to replace the conditional-MLE evaluation point by the reference point. Since $\hat{\alpha}_T - \alpha_T^* = o_p(1)$ and the derivatives of Γ are continuous,

$$\Gamma_\alpha(\hat{\alpha}_T, \beta_T^*; S) - \Gamma_\alpha(\alpha_T^*, \beta_T^*; S) = o_p(1),$$

and

$$\Gamma_{\beta}(\hat{\alpha}_T, \beta_T^*; S) - \Gamma_{\beta}(\alpha_T^*, \beta_T^*; S) = o_p(1).$$

In addition, the Hessian-continuity condition, nonsingularity, and continuity of matrix inversion imply

$$\hat{I}_{\alpha, \alpha}^{-1} \hat{I}_{\alpha, \beta} - \left(\hat{I}_{\alpha, \alpha}^* \right)^{-1} \hat{I}_{\alpha, \beta}^* = o_p(1).$$

Substituting these relations into equation (48) yields

$$\begin{aligned} \left. \frac{\partial \gamma_T(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} &= \Gamma_{\beta}(\alpha_T^*, \beta_T^*; S) \\ &\quad + \Gamma_{\alpha}(\alpha_T^*, \beta_T^*; S) \left(\hat{I}_{\alpha, \alpha}^* \right)^{-1} \hat{I}_{\alpha, \beta}^* + o_p(1) \\ &= \hat{D}_{\gamma, \beta} + o_p(1). \end{aligned}$$

Finally, using the reverse triangle inequality,

$$\begin{aligned} |K_{T, S} - \hat{K}_S| &\leq \sqrt{|S^c|} \left\| \left(\left. \frac{\partial \gamma_T(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} - \hat{D}_{\gamma, \beta} \right) \Sigma_S \right\|_2 \\ &\leq \sqrt{|S^c|} \left\| \left. \frac{\partial \gamma_T(\beta; S)}{\partial \beta'} \right|_{\beta=\beta_T^*} - \hat{D}_{\gamma, \beta} \right\|_2 \|\Sigma_S\|_2 \\ &= o_p(1). \end{aligned}$$

This proves the lemma. □

PROOF OF THEOREM 5. By the asymptotic-equivalence lemma, for every $S \in \mathcal{S}_{AD}$,

$$K_{T, S} - \hat{K}_S = o_p(1).$$

Moreover, by the definition $K_S := \text{plim } \hat{K}_S$,

$$\hat{K}_S - K_S = o_p(1).$$

Since $d_\eta < \infty$, the set $\mathcal{S}_{\text{AD}}$ is finite. Hence both convergences are uniform over $\mathcal{S}_{\text{AD}}$, and therefore

$$\max_{S \in \mathcal{S}_{\text{AD}}} |K_{T,S} - K_S| = o_p(1), \quad \max_{S \in \mathcal{S}_{\text{AD}}} |\hat{K}_S - K_S| = o_p(1).$$

By Assumption 6, K_S has the unique minimizer S^* . Since $\mathcal{S}_{\text{AD}}$ is finite, S^* is separated from every other partition by a strictly positive gap. The uniform convergences above therefore imply

$$\Pr(S_T^* = S^*) \longrightarrow 1, \quad \Pr(\hat{S}^* = S^*) \longrightarrow 1.$$

Consequently,

$$\Pr(S_T^* = \hat{S}^*) \geq 1 - \Pr(S_T^* \neq S^*) - \Pr(\hat{S}^* \neq S^*) \longrightarrow 1.$$

This proves the theorem. □

APPENDIX: REFERENCES

- Al-Sadoon, M. M. and Zwiernik, P. (2019). The identification problem for linear rational expectations models. *arXiv preprint arXiv:1908.09617*. [14]
- Alegre, J. (2026). Choosing what to calibrate and what to estimate in structural models. *arXiv preprint arXiv:2606.25688*. [2, 13, 28, 29]
- An, S. and Schorfheide, F. (2007). Bayesian analysis of dsge models. *Econometric reviews*, 26(2-4):113–172. [2, 14, 15]
- Cocci, M. D. and Plagborg-Møller, M. (2021). Standard errors for calibrated parameters. *arXiv preprint arXiv:2109.08109*. [2]
- Iskrev, N. (2010). Local identification in dsge models. *Journal of Monetary Economics*, 57(2):189–202. [12, 14, 19]
- Iskrev, N. (2019). What to expect when you’re calibrating: Measuring the effect of calibration on the estimation of macroeconomic models. *Journal of Economic Dynamics and Control*, 99:54–81. [3, 7, 11, 19, 28]

- Jørgensen, T. H. (2023). Sensitivity to calibrated parameters. *Review of Economics and Statistics*, 105(2):474–481. [11, 12, 13]
- Kocięcki, A. and Kolasa, M. (2023). A solution to the global identification problem in dsge models. *Journal of Econometrics*, 236(2):105477. [14]
- Komunjer, I. and Ng, S. (2011). Dynamic identification of dynamic stochastic general equilibrium models. *Econometrica*, 79(6):1995–2032. [14]
- Qu, Z. and Tkachenko, D. (2012). Identification and frequency domain quasi-maximum likelihood estimation of linearized dynamic stochastic general equilibrium models. *Quantitative Economics*, 3(1):95–132. [14]
- Qu, Z. and Tkachenko, D. (2017). Global identification in dsge models allowing for indeterminacy. *The Review of Economic Studies*, 84(3):1306–1345. [14]
- Smets, F. and Wouters, R. (2007). Shocks and frictions in us business cycles: A bayesian dsge approach. *American economic review*, 97(3):586–606. [18]